# 上海交通大学课程教学大纲

Carbon Accounting（Global College）
Autumn 2026

<table>
<tr><td colspan="6">课程基本信息（Course Information）</td></tr>
<tr><td>课程代码<br>（Course Code）</td><td>SDG2410J</td><td>*学时<br>（Credit Hours）</td><td>32</td><td>*学分<br>（Credits）</td><td>2</td></tr>
<tr><td rowspan="2">*课程名称<br>（Course Name）</td><td colspan="5">碳核算</td></tr>
<tr><td colspan="5">Carbon Accounting</td></tr>
<tr><td>课程性质<br>(Course Type)</td><td colspan="5">Intellectual Breadth (Other)</td></tr>
<tr><td>授课语言<br>(Language of Instruction)</td><td colspan="5">English</td></tr>
<tr><td>*开课院系<br>（School）</td><td colspan="5">Global College at Shanghai Jiao Tong University</td></tr>
<tr><td>先修课程<br>（Prerequisite）</td><td colspan="5">None</td></tr>
<tr><td>授课教师<br>（Teacher）</td><td colspan="2">Haishan Yu</td><td>课程网址<br>(Course Webpage)</td><td colspan="2"></td></tr>
<tr><td>*课程简介<br>（Description）</td><td colspan="5">Designed for engineering undergraduates, this course examines how greenhouse gas emissions are quantified, reported, and interpreted across different scales and decision contexts. Throughout the course, students apply a common analytical framework—accounting purpose, unit of analysis, boundary, emission attribution, data, and methods—to a range of applications, including national and city inventories, organizational inventories, product carbon footprinting, carbon assessment of project and policy, and financed emissions. The course also considers how accounting results are reported and used, with particular attention to data quality, uncertainty, and credibility.<br>By the end of the course, students will be able to interpret what different carbon accounting results represent and evaluate their applicability and limitations in addressing sustainability issues involving complex engineering systems and technology choices. They will gain experience in conducting simplified carbon accounting for clearly defined cases and</td></tr>
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| | analyzing how choices concerning boundaries, emission attribution, data, and methods affect the results. They will also assess whether accounting results are fit for their intended uses in reporting and decision-making and examine how accounting choices shape the attribution of responsibility and the design of mitigation actions. |
| 课程教学大纲（course syllabus） | |
| *学习目标(Learning Outcomes) | On completion of this course, students will be able to:<br><br>1. **Explain** the main purposes, key concepts, and basic quantification logic of carbon accounting, and distinguish its accounting subjects and principal uses across national, city, organizational, product, project and policy, and financial contexts.<br>2. For a given carbon accounting task, **identify** its purpose and accounting entity, and determine its accounting boundary, emission attribution rules, data requirements, and intended use of the results.<br>3. **Calculate** a simplified greenhouse gas inventory using activity data, emission factors, and appropriate methods, and clearly document data sources and key assumptions.<br>4. **Compare** carbon accounting approaches across scales or contexts, and **analyze** how differences in purpose, accounting boundary, emission attribution, data, and methods affect accounting results and their interpretation.<br>5. **Evaluate** the quality and fitness-for-purpose of carbon accounting results by examining data sources, assumptions, completeness, methodological consistency, uncertainty, and comparability, and determine whether they adequately support the intended reporting, interpretation, or decision-making.<br>6. Collaboratively **design** and iteratively improve an exploratory personal carbon footprint calculator with generative AI support, and individually explain its accounting boundary, emission attribution logic, data sources, assumptions, calculation procedures, major revisions, and limitations. |

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<tr><td>*教学内容、进度安排<br>及要求<br>(Class Schedule<br>& Requirements)</td><td colspan="3"><table>
<tr><td>Lecture 1</td><td>Introduction to Carbon Accounting: course logistics, overview of carbon accounting</td><td>Lecture</td></tr>
<tr><td>Lecture 2</td><td>What is Carbon Accounting and Why Do It</td><td>Lecture</td></tr>
<tr><td>Lecture 3</td><td>How Carbon Accounting Works</td><td>Lecture</td></tr>
<tr><td>Lecture 4</td><td>Attributional Accounting vs Consequential Accounting</td><td>Lecture</td></tr>
<tr><td>Lecture 5</td><td>Carbon Accounting at the national level</td><td>Lecture</td></tr>
<tr><td>Lecture 6</td><td>Carbon Accounting at the community/city level</td><td>Lecture</td></tr>
<tr><td>Lecture 7</td><td>Carbon Accounting at the Firm or Organisational Level_I</td><td>Lecture</td></tr>
<tr><td>Lecture 8</td><td>Carbon Accounting at the Firm or Organisational level_II</td><td>Lecture</td></tr>
<tr><td>Lecture 9</td><td>Product Carbon Footprinting_I</td><td>Lecture</td></tr>
<tr><td>Lecture 10</td><td>Product Carbon Footprinting_II</td><td>Lecture</td></tr>
<tr><td>Lecture 11</td><td>Carbon Accounting at the Project/Policy Level</td><td>Lecture</td></tr>
<tr><td>Lecture 12</td><td>Financed Emissions: Boundaries and Attribution</td><td>Lecture</td></tr>
<tr><td>Lecture 13</td><td>Carbon reporting and disclosure_I</td><td>Lecture</td></tr>
<tr><td>Lecture 14</td><td>Carbon reporting and disclosure_II</td><td>Lecture</td></tr>
<tr><td>Lecture 15</td><td>Project Studio: demonstration</td><td>Lecture</td></tr>
<tr><td>Lecture 16</td><td>Final Exam</td><td>Exam</td></tr>
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<tr><td>*考核方式<br>(Grading)</td><td colspan="3"><ul><li>Attendance and in-class performance (10%)</li><li>In-class quiz (20%)</li><li>Course Project (40%)</li><li>Exam (30%)</li></ul></td></tr>
<tr><td>*教材或参考资料<br>(Textbooks & Other Materials)</td><td colspan="3">This course is NOT a textbook course. The lectures are built upon a series of materials including accounting guidance from various organizations, academic papers, reports and other materials relevant to the course. Most of the materials should be able to be accessed from university library either hardcopy or electronic version.</td></tr>
<tr><td>其它<br>（More）</td><td colspan="3"></td></tr>
<tr><td>备注<br>（Notes）</td><td colspan="3"></td></tr>
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